MIL-R-51279C(EA)
2 September 1976
SUPERSEDING
MIL-R-51279B(MU)
17 September 1973

MILITARY SPECIFICATION

RESUSCITATION TUBE, CHEMICAL-BIOLOGICAL MASK: M1

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a resuscitation tube for use with the M17A1 field chemical-biological mask.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-26 - Bag, Plastic, (General Purpose).
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-26514 - Polyurethane Foam, Rigid or Flexible, for Packaging.

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

FSC 4240

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

D5-81-119 - Marking, Shipping Container for Resuscitation Tube, M1.
D5-81-120 - Palletization, Resuscitation Tube, M1.
D5-81-143 - Resuscitation Tube, Chemical-Biological Mask, M1.
DL136-5-500 - Flow and Leakage Air Tester, Q177.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D5-81-143 and on its subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the resuscitation tube shall conform to the specifications and drawings listed on Drawing D5-81-143 and subsidiary drawings.

3.2 Manufacture and assembly. The resuscitation tube shall be manufactured and assembled to conform to Drawing D5-81-143.

3.3 Assembly separation. The coupling end shall be held at a 45 ± 5° angle and withstand an angular pull of 5.0 + 0.2 pound for 30 ± 5 seconds without any permanent movement of the assembly or damage to any component, when tested as specified in 4.3.3.1. The test shall be repeated with the mouthpiece end.

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3.4 Leakage. The resuscitation tube shall not leak more than 1.5 liters of air per minute at a pressure differential of 1.5 inches of mercury, when tested as specified in 4.4.4.1.

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of resuscitation tubes shall be produced in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship. The resuscitation tube shall be free from contamination and imperfections, such as dirt, grease, oil, foreign matter, damage, porosity, cracks, dents, burrs, sharp edges, surface roughness, and distortion of metal and plastic parts. Any die parting lines which may appear on outside surfaces of any component shall appear as lines only. No flash shall appear on the parting lines. All gates shall be trimmed flush with the molding surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence, acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 25 resuscitation tubes shall be produced, using the same methods, materials, and equipment, as will be used during regular production.

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4.3.2 Inspection procedure. The 25 sample resuscitation tubes shall be examined for all requirements of the drawings and of this specification.

4.3.3 Tests. The 25 sample resuscitation tubes shall be tested in accordance with 4.3.3.1 and 4.4.4.

4.3.3.1 Assembly separation. Resuscitation tube assembly separation shall be determined by using an Instron Tester or similar type of equipment. The machine shall be capable of exerting the required force over the period of time specified in 3.3. The angular pulls applied to mouthpiece and coupling shall be accomplished with universal type adapters capable of being set within the required angular limits. The full cycle of operation should be approximately 1.5 minutes; 30 seconds to apply the required force, 30 seconds dwell time, and 30 seconds to relieve the load. A time tolerance of ± 5 seconds is permissible.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2 and 4.3.3, to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of resuscitation tubes produced by one manufacturer, from the same materials, at the same plant, and under essentially the same manufacturing conditions. However, no more than one lot of rubber components shall be represented in any one lot of finished resuscitation tubes. Any change in material, equipment, or procedure used in regular production which differs from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling for examination and tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure. The sample resuscitation tubes shall be inspected in accordance with the classification of defects and with MIL-STD-105.

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4.4.3.1 Classification of defects.

(a) Resuscitation tube (Drawing D5-81-143).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 0.10 percent defective	
101	Leakage	4.4.4.1
	AQL 1.0 percent defective	
102	Part missing or incorrectly assembled	
103	Workmanship (3.6)	

(b) Preparation for delivery (section 5).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 2.5 percent defective	
101	Cleaning inadequate	5.1
102	Sealing of plastic bag incorrect	5.2.1
103	Fiberboard container incorrect	5.2.1
104	Packaging assembly incorrect	5.2
105	Shipping container incorrect	5.3
106	Packing assembly incorrect	5.3
107	Strapping or closure incorrect	5.3
108	Marking incorrect, missing, or illegible	5.4
109	Palletization (marking, strapping, or pallet) incorrect	5.5

4.4.4 Tests.

4.4.4.1 Leakage. The leakage of the resuscitation tubes shall be determined, using Edgewood Arsenal Q177 Airflow and Leakage Tester (DL136-5-500).

5. PREPARATION FOR DELIVERY

5.1 Cleaning. The resuscitation tubes shall be dry and free of dirt and other foreign matter when packaged.

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5.2 Packaging. Level shall be as specified; see 6.2.

5.2.1 Level A. The resuscitation tubes shall be heat sealed, one per bag (unit package), within polyethylene bags conforming to type II, style 1 of PPP-B-26, having dimensions of 5 inches wide by 14 inches long. The included air volume inside the bag shall be kept to a practicable minimum. NOTE: Do not compress the "O" ring by tightening the adjusting screw of the coupling, prior to packaging. The packaged resuscitation tubes shall be further packaged, 10 per box (intermediate package), in fiberboard boxes conforming to style RSC, grade W6c of PPP-B-636, having inside dimensions of 12 by 8 by 4 inches (length, width, and depth respectively). Arrangement of the tubes in the box shall be lengthwise, alternating the mouthpiece end to best utilize spzce within the box and produce a compact assembly. A sheet of polyurethane cushioning material, approximately 1 inch thick by 12 by 8 inches conforming to type I, class 2, density 1 to 1 1/2 pounds per cubic foot, of MIL-P-26514 shall be included to restrict free movement of the tubes. The box shall be closed in accordance with the method III closure requirements of the appendix to PPP-B-636.

5.2.2 Level B. Packaging shall be in accordance with 5.2.1 except that the boxes shall conform to domestic class, grade 175, type CF, of PPP-B-636 and the closure of the boxes shall be in accordance with either the method I, II, or IV closure requirement of the appendix to PPP-B-636.

5.2.3 Commercial packaging. The resuscitation tubes shall be individually packaged in accordance with FED-STD-356. Note: Do not compress the 'O'-ring (by tightening the adjusting screw of the coupling) prior to packaging.

5.3 Packing. Level shall be as specified; see 6.2.

5.3.1 Level A. The resuscitation tubes, packaged in accordance with 5.2, shall be packed, 40 (four packages) per box, in nailed wood boxes conforming to style 4, class 2, grade A, 0- to 50-pound limit for type 1 or 2 load for overseas shipment of PPP-B-621, having inside dimensions of 18 1/2 by 12 1/2 by 8 1/2 inches (length, width, and depth, respectively). The boxes shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to class A or B, type I or IV, of QQ-S-781.

5.3.2 Level B. The resuscitation tubes, packaged as specified in the contract or order, shall be packed, 40 (four packages) per box; in fiberboard boxes conforming to style RSC, grade V3c, of PPP-B-636, having inside dimensions of 18 1/2 by 12 1/2 by 8 1/2 inches (length, width, and depth, respectively). The boxes shall be closed in accordance with the method III closure requirements of the appendix to PPP-B-636.

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5.3.3 Commercial packing. Packaged resuscitation tubes shall be packed in accordance with FED-STD-356.

5.4 Marking. In addition to marking required by the contract or order, marking shall be as specified herein.

5.4.1 Unit packages (5.2). A label containing the National Stock Number, item nomenclature, and contract or purchase order number shall be sealed within the package and positioned so that the markings are visible from the outside of the package. The marking shall be in accordance with MIL-STD-129 for unit packaging.

5.4.2 Intermediate packages (5.2). The packages shall be marked with the National Stock Number, item nomenclature, quantity and unit of issue, contract or purchase order number, level of packaging, and date packaged. The marking shall be in accordance with MIL-STD-129 for intermediate packaging.

5.4.3 Packing marking. The shipping container (5.3) shall be marked in accordance with Drawing D5-81-119.

5.4.4 Commercial marking. Marking of commercial packages and shipping containers shall be in accordance with FED-STD-356.

5.5 Palletization. The resuscitation tubes, packed in accordance with 5.3, shall be palletized in accordance with Drawing D5-81-120. Commercial unit packs shall be palletized at the discretion of the contractor or when specified in the contract or order to meet carrier acceptance and provide safe delivery to destination at lowest rates in compliance with the carrier rules applicable to the mode of shipment.

6. NOTES

6.1 Intended use.

6.1.1 Resuscitation tube. The resuscitation tube covered by this specification is intended to be used with the M17A1 Field Chemical-Biological Mask.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packing and packaging.
(c) Palletization.
(d) Preproduction.

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(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A683

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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